

Abhijit Shinde

PhD Candidate | Electrical and Computer Engineering | North Carolina A&T State University

Contact

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Achievements

- NASA Reviewer: MUREP Aeronautics Community Engagement Impact Research (ACEIR) 2.0
- Finalist – NIST UAS 5.0 Challenge for autonomous drone mapping and navigation
- Winner – 1st Round of GoAero (Aerial Emergency Response Operations) Competition

Skills

RF & Microwave Engineering: HFSS, CST Studio Suite, ADS, XFDTD, VNA Calibration, S-Parameters, Return Loss, VSWR, Beamforming, Radiation Pattern Analysis, Filter Design, Impedance Matching

Antenna Design & Simulation: Vivaldi, Patch, and Slot Antennas; UWB and Microstrip Structures; SAR and Near-field Simulations

Programming & Analysis Tools: MATLAB, Python, Simulink, TensorFlow, Scikit-learn, Pandas, NumPy

Measurement & Testing: Vector Network Analyzer (VNA), Spectrum Analyzer, Anechoic Chamber Testing, PCB Prototyping

Certifications: Six Sigma Green Belt | NVIDIA DLI Deep Learning | MathWorks ML/DL

Education

- **Ph.D., Electrical and Computer Engineering** – North Carolina A&T State University (2023– Expected 2027) | GPA: 3.68/4.0
- **M.E. in Electronics and Telecommunication** – Pune University (2014–2016) | GPA: 3.5/4.0
- **B.E. in Electronics and Telecommunication** – Shivaji University (2004–2007) | GPA: 3.29/4.0

Publications

Total Publications: 5 Citations: 10

Key Publication: A. Shinde et al., “Blood spatter trajectory analysis for spatter source reconstruction using image processing,” IEEE CASP 2016 Full list:

[Google Scholar](#)



Professional Summary

Ph.D. Candidate in Electrical and Computer Engineering at North Carolina A&T State University with expertise in RF, microwave, and antenna design for advanced sensing and communication systems. Skilled in HFSS, CST, ADS, XFDTD, and MATLAB for the design and optimization of tunable filters, broadband antennas, and UWB imaging modules. Proven ability to bridge simulation and experimental validation, delivering high-performance RF prototypes for biomedical and aerospace applications. Honored as a NASA MUREP Reviewer, NIST UAS Finalist, and GoAero Award Winner.



Professional Experience

PhD Researcher – North Carolina A&T State University (2023–2027)

- Designed and simulated UWB Vivaldi antennas in HFSS achieving $S_{11} < -10$ dB (1–6 GHz).
- Developed microwave imaging system for brain hemorrhage detection with >95% CNN prediction accuracy.
- Investigated YIG-based tunable filter structures with optimized coupling and out-of-band attenuation.
- Collaborated with NASA ULI team on hybrid UAV communication systems integrating microwave sensors for aerial navigation.
- Performed SAR analysis and electromagnetic safety evaluation on six-layer human head models.

Assistant Professor – Sinhgad Institute of Technology (2022–2023)

Taught **RF Circuit Design**, **Data Analytics**, and **Digital Electronics**; mentored student antenna design projects improving results by 35%.

Assistant Professor – Dr. D. Y. Patil Institute of Technology (2017–2022)

Delivered courses in Digital Signal Processing, Image Processing, and Electromagnetics.

Automation Engineer – KalpTech Solutions Pvt. Ltd. (2014–2016)

Automated PCB inspection using ML, reducing manual testing by 95%.

Automation Engineer – Rapid Pack Engineering Pvt. Ltd. (2009–2010)

Programmed industrial machines for pharmaceutical packaging.

Lecturer – KIT's College of Engineering (2008–2009)

Courses Taught: Microprocessors, Computer Networking, Digital Electronics.



Key Projects

YIG-Tunable Filter Design (HFSS & ADS):

Developed planar tunable filter achieving >20 dB stopband attenuation and <3 dB insertion loss; analyzed coupling variation under magnetic bias.

UWB Vivaldi Antenna Array:

Designed a broadband Vivaldi array (1–7 GHz) for microwave imaging; optimized gain and directivity using exponential taper geometry.

Microwave Imaging for Intracranial Hemorrhage Detection:

Integrated antenna array and CNN-based regression model to localize hemorrhage regions; validated with phantom models.

Sensor-Based Microwave Brain Imaging System (SMBIS):

Built a six-layer head phantom to evaluate SAR and dielectric properties using broadband measurements.